



DM Unit-1 pdf

Discrete Mathematics (Jawaharlal Nehru Technological University, Hyderabad)



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UNIT-1

Mathematical Logic: It is the logic of symbols with using a large statement.

$\wedge$ - and

$\vee$ - or

$\sim$ - negation

$\rightarrow$ - implies

$\Rightarrow$ - double implies

$\forall$ - for all

$\circ$ - contradiction

$\exists$ - there exist

$\in$ - belongs to

$\notin$ - does not belongs to

$\vdash$ - such that

$\equiv$ - logically equivalence

T - tautology

$-$ - Complement

Compound statements: Compound statements are AND, OR, Negation, implies, double implies.

And - It is one of the Compound statement, which is used to linking statement. AND is a con

Truth table: P and Q both are true - output is true
if any one is false - output is false both are false
- Output is false

P	Q	$P \wedge Q$	Output
T	T	T	1
T	F	F	0
F	T	F	0
F	F	F	0

Ex: Radha and Krishna go to movie

Radha = P

Krishna = Q

$P \wedge Q$

OR - It is one of the compound statement if any 1 is true output is both are true

• If 1 is true and one is false both are true

• Both are false then output is false

Truth table :

P	q	$P \vee q$	output	example
T	T	T	1	$1+1=1$
T	F	T	1	$1+0=1$
F	T	T	1	$0+1=1$
F	F	F	0	$0+0=0$

Negation: It is one of the compound statement the statement is the -ve statement, -ve statement is the

Truth table :

P is true negation of P is false
P is false negation of P is true

P	$\neg P$	output
T	F	0
F	T	1

Ex: $\neg(1) = 0$
 $\neg(0) = 1$

Implies: ($\rightarrow$) It is one of the compound statement if P and Q both are true then output is true

If P is true and Q is false output is true

If P is false and Q is true output is false

Both are false then output is true

Truth table :

P	q	$P \rightarrow q$	output
T	T	T	1
T	F	F	0
F	T	T	1
F	F	T	1

double implies: ($\leftrightarrow$) It is one of the compound statement if P and Q both are true then output is true
If anyone is false output is false
Both are false, output is true

Truthtable:

P	q	$P \leftrightarrow q$	output
T	T	T	1
T	F	F	0
F	T	F	0
F	F	T	1

Note: Compound statements are also said to be preposition statements.

Problems:-

1. And $P \wedge (\neg q \vee P)$ by using truth table

P	q	$\neg q$	$(\neg q \vee P)$	$P \wedge (\neg q \vee P)$	output
T	T	F	T	T	1
T	F	T	T	T	1
F	T	F	F	F	0
F	F	T	F	F	0

2. By using truthtable:

- I. $[\neg P \wedge (q \vee P)] \leftrightarrow P$
- II. $[(\neg P) \vee (\neg q)] \wedge [(\neg P) \vee r]$
- III. $[P \rightarrow (q \rightarrow r)] \rightarrow [(P \wedge q) \vee r]$
- IV. $(P \wedge q) \vee [(\neg P) \rightarrow q]$

1. $[\neg P \wedge (q \vee P)] \leftrightarrow P$

P	q	$\neg P$	$q \vee P$	$(\neg P \wedge q \vee P)$	$[\neg P \wedge (q \vee P)] \leftrightarrow P$	output
T	T	F	T	F	F	0
T	F	F	T	F	F	0
F	T	T	T	T	F	0
F	F	T	F	F	T	1

II. $[(\neg p) \vee (\neg q)] \wedge [(\neg p) \vee r]$

P	q	r	$\neg p$	$\neg q$	$(\neg p \vee \neg q)$	$(\neg p \vee r)$	$[(\neg p) \vee (\neg q)] \wedge [(\neg p) \vee r]$	output
T	T	T	F	F	F	T	F	0
T	T	F	F	F	F	F	F	0
T	F	T	F	T	T	T	T	1
T	F	F	F	T	T	F	F	0
F	T	T	T	F	T	T	T	1
F	T	F	T	F	T	T	T	1
F	F	T	T	T	T	T	T	1
F	F	F	T	T	T	T	T	1

III. $(P \rightarrow (q \rightarrow r)) \rightarrow [(P \wedge q) \vee r]$

P	q	r	$q \rightarrow r$	$P \rightarrow (q \rightarrow r)$	$(P \wedge q)$	$(P \wedge q) \vee r$	Full/total	output
T	T	T	T	T	T	T	T	1
T	T	F	F	F	T	F	F	0
T	F	T	T	T	F	T	T	1
T	F	F	T	T	F	F	F	0
F	T	T	T	T	F	T	T	1
F	T	F	F	F	F	F	F	0
F	F	T	T	T	F	T	T	1
F	F	F	T	T	F	F	F	0

IV. $(P \wedge q) \vee [(\neg p) \rightarrow q]$

P	q	$P \wedge q$	$\neg p$	$(\neg p \rightarrow q)$	$(P \wedge q) \vee [(\neg p) \rightarrow q]$	output
T	T	T	F	F	T	1
T	F	F	F	T	T	1
F	T	F	T	T	T	1
F	F	F	T	F	F	0

Logical equivalence:

→ A large statement we can assign the symbol (logical equivalence) to the LHS = RHS

$$\neg(P \vee q) \equiv (\neg P) \wedge (\neg q)$$

$$\neg(P \wedge q) \equiv (\neg P) \vee (\neg q)$$

$$P \vee T \equiv T$$

$$P \wedge F \equiv F$$

$$P \rightarrow q \equiv \neg P \vee q$$

P: Prove that $\neg(P \vee q) \equiv (\neg P) \wedge (\neg q)$ is logically equivalence

P	q	$(P \vee q)$	$\neg(P \vee q)$	$\neg P$	$\neg q$	$(\neg P) \wedge (\neg q)$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

∴ It is logically equivalence to the
 $\neg(P \vee q) \equiv (\neg P) \wedge (\neg q)$ hence proved

② $\neg(P \wedge q) \equiv (\neg P) \vee (\neg q)$

P	q	$(P \wedge q)$	$\neg(P \wedge q)$	$\neg P$	$\neg q$	$(\neg P) \vee (\neg q)$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

∴ It is logically equivalence to the

$$\neg(P \wedge q) \equiv (\neg P) \vee (\neg q)$$

$$\textcircled{3} P \vee T \equiv T$$

P	T	$P \vee T$	T
T	T	T	T
T	T	T	T
F	T	T	T
F	T	T	T

$$\therefore P \vee T \equiv T$$

$$\textcircled{4} P \wedge F \equiv F$$

P	F	$P \wedge F$	F
T	F	F	F
T	F	F	F
F	F	F	F
F	F	F	F

$$\therefore P \wedge F \equiv F$$

$$\textcircled{5} P \rightarrow q \equiv \sim p \vee q$$

P	q	$P \rightarrow q$	$\sim p$	$\sim p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$\therefore$ It is not logically equivalent.

Mathematical Expressions

① $P \wedge T \equiv P$
 $P \vee F \equiv P$ } Identity law

② $P \vee T \equiv T$
 $P \wedge F \equiv F$ } Domination law

③ $(P \vee Q) \equiv (Q \vee P)$
 $(P \wedge Q) \equiv (Q \wedge P)$ } Commutative law

④ $P \vee (Q \vee R) \equiv (P \vee Q) \vee R$
 $P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$ } Associative law

⑤ $\neg(\neg P) \equiv P$ } double negation law

⑥ $P \vee \neg P \equiv T$
 $P \wedge \neg P \equiv F$ } negation law

⑦ $P \vee (P \wedge R) \equiv (P \vee R)$
 $P \wedge (P \vee R) \equiv (P \wedge R)$ } Idempotent law

⑧ $\neg(P \vee Q) \equiv (\neg P) \wedge (\neg Q)$
 $\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$ } Demorgan law

① $P \wedge T \equiv P$

② $P \vee F \equiv P$

P	T	$P \wedge T$	P
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F

$$\textcircled{3} (p \vee q) \equiv (q \vee p)$$

P	q	$p \vee q$	$q \vee p$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

$$(p \wedge q) \equiv (q \wedge p)$$

P	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

$$\textcircled{4} p \vee (q \vee r) \equiv (p \vee q) \vee r$$

P	q	r	$q \vee r$	$p \vee (q \vee r)$	$(p \vee q) \vee r$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	T	T	T
T	F	F	F	T	T
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	T	T
F	F	F	F	F	F

$$\therefore p \vee (q \vee r) \equiv (p \vee q) \vee r$$

$$P \wedge (Q \wedge R) \equiv (P \wedge Q) \wedge R$$

P	Q	R	$Q \wedge R$	$P \wedge (Q \wedge R)$	$(P \wedge Q)$	$(P \wedge Q) \wedge R$
T	T	T	T	T	T	T
T	T	F	F	F	T	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	T	F	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

$$\textcircled{3} \neg(\neg P) \equiv P$$

P	$\neg P$	$\neg(\neg P)$	P
T	F	T	T
T	F	T	T
F	T	F	F
F	T	F	F

$$\textcircled{6} P \vee \neg P \equiv T$$

P	$\neg P$	$P \vee \neg P$	T
T	F	T	T
T	F	T	T
F	T	T	T
F	T	T	T

$$P \wedge \neg P \equiv F$$

P	$\neg P$	$P \wedge \neg P$	F
T	F	F	F
T	F	F	F
F	T	F	F
F	T	F	F

$$\textcircled{7} P \vee (q \wedge r) \equiv (P \vee q) \wedge (P \vee r)$$

P	q	r	$q \wedge r$	$P \vee (q \wedge r)$	$P \vee q$	$P \vee r$	$(P \vee q) \wedge (P \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

$$P \wedge (q \vee r) \equiv (P \wedge q) \vee (P \wedge r)$$

P	q	r	$q \vee r$	$P \wedge (q \vee r)$	$P \wedge q$	$P \wedge r$	$(P \wedge q) \vee (P \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

$$\textcircled{8} \sim(P \vee q) \equiv (\sim P) \wedge (\sim q)$$

P	q	$P \vee q$	$\sim(P \vee q)$	$\sim P$	$\sim q$	$(\sim P) \wedge (\sim q)$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

$$\neg(P \wedge Q) \equiv (\neg P) \vee (\neg Q)$$

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$(\neg P)$	$(\neg Q)$	$(\neg P) \vee (\neg Q)$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

* Power set: If it is set of all subset to the main set it is denoted by $P(n) = 2^n$

$$\text{ex: } A = \{1, 2, 3\}$$

$$P(A) = \{ \{\emptyset\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

Theorems

1. show that $(\neg p) \rightarrow (p \rightarrow q)$ is a tautology by logical expression (without truth table)

$$\text{Proof: } (\neg p) \rightarrow (p \rightarrow q)$$

$$\equiv \neg p \rightarrow (\neg p \vee q)$$

$$\equiv p \vee q$$

$$\equiv p \vee (\neg p \vee q)$$

$$\equiv (p \vee (\neg p)) \vee q$$

$$\equiv T \vee q$$

$$\equiv T$$

$$[(\neg p) \rightarrow q \equiv p \vee q]$$

$$[p \rightarrow q \equiv \neg p \vee q]$$

$$q \equiv \neg p \vee q$$

($\therefore$ By associative law)

2. show that negation of $p \vee (\neg p) \wedge q$ are logically equivalence by $\neg p \wedge \neg q$ (without truth table)

LHS

$$p \vee (\neg p) \wedge q$$

$$\neg[p \vee (\neg p) \wedge q]$$

$$\equiv \neg p \wedge (\neg(\neg p) \vee (\neg q)) \quad (\therefore \text{double negation})$$

$$\equiv \neg p \wedge (p \vee \neg q) \quad (\text{By Distributive law})$$

$$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$$

$$\equiv F \vee (\neg p \wedge \neg q)$$

$$\equiv (\neg p \wedge \neg q)$$

3. Show that $(P \wedge Q) \rightarrow (P \vee Q)$ is a tautology by using logically equivalence.

$$(P \wedge Q) \rightarrow (P \vee Q)$$

$$[P \rightarrow Q \equiv \neg P \vee Q]$$

$$\equiv \neg P \vee Q$$

$$\equiv \neg(P \wedge Q) \vee (P \vee Q)$$

$$\equiv \neg P \vee \neg Q \vee (P \vee Q)$$

$$\equiv (\neg P \vee P) \vee (\neg Q \vee Q)$$

$$\equiv T \vee T$$

$$\equiv T$$

$\therefore$ It is logically equivalence to the tautology.

4. Show that $P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$ by using truth table to the logically equivalence.

$$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$$

P	Q	$P \leftrightarrow Q$	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$
T	T	T	T	T	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

$\therefore$ It is logically

5. Show that negation $[\neg(P \wedge (P \vee Q))] \rightarrow Q$ is tautology by using logical equivalence (without truth table)

$$[\neg(P \wedge (P \vee Q))] \rightarrow Q$$

$$\equiv [(\neg P \wedge P) \vee (\neg P \wedge Q)] \rightarrow Q \quad [\text{Distributive law}]$$

$$\equiv [F \vee (\neg P \wedge Q)] \rightarrow Q \quad [\because \text{Negation law}]$$

$$\equiv [(F \vee \neg P) \wedge (F \vee Q)] \rightarrow Q$$

$$\equiv [\neg P \wedge Q] \rightarrow Q \quad (\because \text{Identity law})$$

$$\equiv \neg(\neg P \wedge Q) \rightarrow Q$$

$$\equiv \sim(\sim p) \vee \sim q \vee q$$

$$\equiv p \vee (\sim q \vee q) \quad (\text{Associative Law})$$

$$\equiv p \vee T \quad (\text{Domination Law})$$

$$\equiv T$$

6. Show that $(p \wedge q) \rightarrow p$ is a tautology by using logical equivalence

$$(p \wedge q) \rightarrow p$$

$$p \rightarrow q$$

$$\equiv \sim p \vee q$$

$$\equiv \sim(p \wedge q) \vee p$$

$$\equiv \sim p \vee \sim q \vee p$$

$$\equiv \sim p \vee p \vee \sim q$$

$$\equiv (\sim p \vee p) \vee \sim q$$

$$\equiv T \vee \sim q$$

$$\equiv T$$

$$(\sim p) \rightarrow q \equiv p \vee q$$

$$p \rightarrow q \equiv \sim p \vee q$$

7. Show that $\sim[p \vee (\sim p \wedge q)]$ is logically equivalence to the $\sim p \wedge \sim q$

$$\sim[p \vee (\sim p \wedge q)] \equiv \sim p \wedge \sim q$$

$$\equiv \sim[p \vee (\sim p \wedge q)] \quad (\text{by using Negation law})$$

$$\equiv \sim p \wedge (\sim(\sim p \wedge q))$$

$$\equiv \sim p \wedge (p \vee \sim q) \quad (\text{by Distributive law})$$

$$(\sim p \wedge p) \vee (\sim p \wedge \sim q)$$

$$F \vee (\sim p \wedge \sim q)$$

$$\sim p \wedge \sim q$$

1) show that $[(p \vee q) \vee (q \vee (\neg r) \wedge (p \vee r))] \equiv \neg[(\neg p \wedge (\neg q))]$
by using truth table

2) $p \leftrightarrow q \equiv (\neg p) \leftrightarrow (\neg q)$

3) $\neg(p \leftrightarrow q) \equiv p \leftrightarrow (\neg q)$

4) $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

5) $p \rightarrow q \equiv (\neg p) \vee q$

6) $\neg(p \vee q) \leftrightarrow [(\neg p) \wedge (\neg q)]$ is a tautology by using truth table

1. $[(p \vee q) \vee (q \vee (\neg r) \wedge (p \vee r))] \equiv \neg[(\neg p \wedge (\neg q))]$

p	q	r	$p \vee q$	$\neg r$	$q \vee (\neg r)$	$p \vee r$	$[(q \vee (\neg r)) \wedge (p \vee r)]$	$[(p \vee q) \vee (q \vee (\neg r) \wedge (p \vee r))]$	$\neg p$	$\neg q$	$\neg(\neg p \wedge \neg q)$
T	T	T	T	F	T	T	T	T	F	F	F
T	T	F	T	T	T	T	T	T	F	F	F
T	F	T	T	F	F	T	F	T	F	T	F
T	F	F	T	T	T	T	T	T	F	T	F
F	T	T	T	F	T	T	T	T	T	F	F
F	T	F	T	T	T	F	F	T	T	F	F
F	F	T	F	F	F	T	F	T	T	F	F
F	F	F	F	T	T	F	F	F	T	T	T

2. $p \leftrightarrow q \equiv (\neg p) \leftrightarrow (\neg q)$

p	q	$p \leftrightarrow q$	$\neg p$	$\neg q$	$(\neg p) \leftrightarrow (\neg q)$
T	T	T	F	F	T
T	F	F	F	T	F
F	T	F	T	F	F
F	F	T	T	T	T

3. $\neg(p \leftrightarrow q) \equiv p \leftrightarrow (\neg q)$

p	q	$p \leftrightarrow q$	$\neg(p \leftrightarrow q)$	$\neg q$	$p \leftrightarrow (\neg q)$
T	T	T	F	F	F
T	F	F	T	T	T
F	T	F	T	F	T
F	F	T	F	T	F

4. $(P \leftrightarrow Q) \equiv (P \wedge Q) \vee (\neg P \wedge \neg Q)$

P	Q	$P \leftrightarrow Q$	$P \wedge Q$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$
T	T	T	T	F	F	F	T
T	F	F	F	F	T	F	F
F	T	F	F	T	F	F	F
F	F	T	F	T	T	T	T

5. $P \rightarrow Q \equiv (\neg P) \vee Q$

P	Q	$P \rightarrow Q$	$\neg P$	$(\neg P) \vee Q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

6. $\neg(P \vee Q) \Leftrightarrow ((\neg P) \wedge (\neg Q))$

P	Q	$P \vee Q$	$\neg P$	$\neg Q$	$(\neg P) \wedge (\neg Q)$	$\neg(P \vee Q) \Leftrightarrow ((\neg P) \wedge (\neg Q))$
T	T	T	F	F	F	T
T	F	T	F	T	F	T
F	T	T	T	F	F	T
F	F	F	T	T	T	T

Predicates:-

* A statement or a relationship to an object is neither true or false

Conditions:- $\nexists x > 3$

$\nexists x = y + 3$

$\exists x + y = z$

It is denoted by $P(x)$ x is subject of the statement applied to the condition is in real numbers $x \in R$

Q) what is the truth value of the predicate $P(x)$ denotes the statement $x > 3$ for $x = 4$ & $x = 2$

Sol:- $P(2)$ for $x = 2$ $2 > 3$ false

$P(4)$ for $x = 4$ $4 > 3$ True

$\therefore$ It is a predicate of $x \in R$

Q) let $\phi(x,y)$ denotes the statement $x=y+3$ what are truth values of proposition $\phi(1,2), \phi(3,0)$

$$x=y+3$$

$$\phi(x,y) \Rightarrow x=1 \text{ \& } y=2 \Rightarrow 1=2+3 \rightarrow \text{false}$$

$$\phi(x,y) \Rightarrow x=3 \text{ \& } y=0 \Rightarrow 3=0+3 \rightarrow \text{true}$$

$\therefore$ It is a predicate of $x \in \mathbb{R}$

Q) what are the truth value of proposition $R(1,2,3)$ and $R(0,0,1)$ where $R(x,y,z)$ denote the statement of $x+y=z$

$$x+y=z$$

$$x+y=z$$

$$0+0=1 \rightarrow \text{False}$$

$$1+2=3 \rightarrow \text{True}$$

Quantifiers:- They are 3 types of quantifiers

i) Universal Quantifier:- we can use logically (for all $\forall$) is called Universal Quantifier the universal quantifier of the statement $P(x)$ $\forall x$ in the given domain. The universal quantifier is denoted by $\forall x P(x)$

Ex:- let $P(x)$ be the statement $x+1 > x$ what is the truth value of quantification $\forall x P(x)$

Sol:- Given statement is $x+1 > x$

Universal quantifier is $\forall x P(x)$

Given domain is $\mathbb{P}$

$\therefore \forall x P(x)$ is true

Note:- x is a real values of all elements $x_1, x_2, x_3, \dots, x_n$ of the domain $P(x_1), P(x_2), \dots, P(x_n)$ is same as the universal Quantifier.

* Universal quantifier is true for the conjunction if and only if all real values of x is true for the domain $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$

Q) what are the truth value of $\forall x P(x)$ $P(x)$ is the statement $x^2 > 10$ what are the real values consistent in a domain not exceeding for 5.

Sol:- Given statement $x^2 > 10$

Quantifier = $\forall x P(x)$

domain is $P(x)$

let us take $\{1, 2, 3, 4, 5\}$

$\downarrow$
Because 'x'

$$P(1) : 1^2 > 10 \text{ F}$$

$$P(2) : 2^2 > 10 \text{ F}$$

$$P(3) : 3^2 > 10 \text{ F}$$

$$P(4) : 4^2 > 10 \text{ T}$$

$$P(5) : 5^2 > 10 \text{ T}$$

$$P(1) \wedge P(2) \wedge P(3) \wedge P(4) \wedge P(5)$$

$$\rightarrow F \wedge F \wedge F \wedge T \wedge T$$

$$\rightarrow F$$

$\forall x$ is a real value is false

Q) what are the truth values $\forall x P(x)$. $P(x)$ is a statement $2x > x$. what are the real values of an integer not exceeding for 4

Sol: given statement, $2x > x$

Quantification = $\forall x P(x)$

domain is $P(x)$

$$P(1) : 2(1) > 1 \rightarrow T$$

$$P(2) : 2(2) > 2 \rightarrow T$$

$$P(3) : 2(3) > 3 \rightarrow T$$

$$P(4) : 2(4) > 4 \rightarrow T$$

Essential quantifiers: $(\exists)$:

* It is a essential quantifiers we can assign the symbol $\exists$ if it exists.

* 'x' is a real value of domain $P(x)$ then $\exists x P(x)$

Ex: Sum, few, many etc

Note: Let 'x' is a real value of all elements $x_1, x_2, \dots, x_n$ to the domain $P(x_1), P(x_2), \dots, P(x_n)$ is same as the essential quantifiers

* Essential quantifier is for disjunction if and only if at least one is true for the statement of domain
 $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$.

Q) what is the truth value of $\exists x P(x)$. $P(x)$ is the statement $x^2 + x - 9 < 25$ and domain the set of all positive integers not exceeding 5

Sol: Given statement $x^2 + x - 9 < 25$

essential quantifier $= \exists x P(x)$

domain is $P(x)$

$$P(1): (1)^2 + 1 - 9 < 25 \quad T$$

$$P(2): 4 + 2 - 9 < 25 \quad T$$

$$P(3): 9 + 3 - 9 < 25 \quad T$$

$$P(4): 16 + 4 - 9 < 25 \quad T$$

$$P(5): 25 + 5 - 9 < 25 \quad T$$

$$\Rightarrow P(1) \vee P(2) \vee P(3) \vee P(4) \vee P(5)$$

$$\Rightarrow T \vee T \vee T \vee T \vee T$$

$$\Rightarrow T$$

$\therefore \exists x P(x)$ is a real value is True

Q) Let $P(x)$ is denotes the statement $x > 3$ what is the value of quantification $\exists x P(x)$. where the domain is the set of real number.

Nested quantifiers:-

* Universal and existential quantifiers are both quantification is said to be Nested quantifiers.

Eg: $\forall x P(x)$ & $\exists x P(x)$

Q: Let $P(x)$ be the statement $x = x^2$. If the domain consists of integers what are the truth values of $P(0)$, $P(1)$, $P(2)$, $P(-1)$ then find $\forall x P(x)$ & $\exists x P(x)$

Ans: Given statement $x = x^2$

Quantification $\forall x P(x)$

domain is $P(x)$

$$P(0) : 0 = 0 \quad T$$

$$P(1) : 1 = 1 \quad T$$

$$P(2) : 2 = 4 \quad F$$

$$P(-1) : -1 = 1 \quad F$$

$$\Rightarrow P(0) \wedge P(1) \wedge P(2) \wedge P(-1)$$

$$\Rightarrow T \wedge T \wedge F \wedge F$$

$$\Rightarrow F$$

$\therefore \forall x P(x)$ is a real value is false

Given statement $x = x^2$

Quantification $\exists x P(x)$

domain is $P(x)$

$$P(0) : 0 = 0 \quad T$$

$$P(1) : 1 = 1 \quad T$$

$$P(2) : 2 = 4 \quad F$$

$$P(-1) : -1 = 1 \quad F$$

$$\Rightarrow P(0) \vee P(1) \vee P(2) \vee P(-1)$$

$$\Rightarrow T \vee T \vee F \vee F$$

$$\Rightarrow T$$

$\therefore \exists x P(x)$ is a real value true

Q) Determine the truth value of following statement. If the domain consistent of all the real numbers.

i) $\exists x (x^3 = -1)$

ii) $\forall x ((1-x)^2 = x^2)$

iii) $\exists x (x^4 \leq x^2)$

iv) $\forall x (2x > x)$

i) $\exists x (x^3 = -1)$

Sol: $x^3 = -1 \rightarrow$ domain

for $x = -1$

$(-1)^3 = -1$

$\therefore \exists x p(x)$ the domain is exist true for some $x = -1$

ii) $\forall x ((1-x)^2 = x^2)$

Sol: $(1-x)^2 = x^2$

for $x = 1$

$(1-1)^2 = 1^2$

$\therefore \forall x p(x)$ is exist true for all $x = 1$

iii) $\exists x (x^4 \leq x^2)$

Sol: for $x = 1$

$1^4 \leq 1^2 \quad T$

for $x = -1$

$(-1)^4 \leq (-1)^2 \quad T$

$\therefore \exists x p(x)$ is true some real numbers.

iv) $\forall x (2x > x)$

Sol: $\forall x p(x)$

$2x > x$

$P(1) : 2 > 1 \quad T$

$P(2) : 4 > 2 \quad T$

$\therefore \forall x p(x)$ is true for all real value.

Duality:-

The complimentary of Quantification is said to be duality it is used to the Quantification of the negation of the statement.

Ex:- All birds are flying in the sky

$$\neg \neg = \neg$$

$$\neg \neg = \neg$$

$$\neg \neg = \neg$$

$$\neg \neg = \neg$$

$$\neg \neg = \neg$$

$$\neg \neg = \neg$$

$$\neg \neg = \neg$$

$$\forall x p(x)$$

$$\neg [\forall x p(x)]$$

$$\exists x \neg p(x)$$

Express the statement every student in this class has studied calculus using predicates and Quantifiers.

Sol:- Every student in this class

$$\forall x s(x), c(x)$$

$$\forall x [s(x) \rightarrow c(x)]$$

$$\neg [\forall x [s(x) \rightarrow c(x)]]$$

$$\exists x [\neg s(x) \vee (\neg c(x))]$$

$$\exists x \neg [s(x) \wedge c(x)]$$

All students visited in this class visited Mexico and Every student in this class have visited either Canada or Mexico.

Student in the class $s(x)$

visited Mexico = $m(x)$

Canada = $c(x)$

$$\forall x s(x), \forall x m(x), \forall x c(x)$$

$$\forall x s(x) \rightarrow \forall x m(x)$$

$$\forall x [s(x) \rightarrow m(x)]$$

$$\neg \forall x [s(x) \rightarrow m(x)]$$

$$\exists x [\neg s(x) \vee \neg m(x)]$$

$$\exists x \neg [s(x) \wedge m(x)]$$

$$\forall x s(x) \rightarrow \forall x m(x) \vee \forall x c(x)$$

$$\neg \forall x [s(x) \rightarrow m(x) \vee c(x)]$$

$$\exists x [s(x) \wedge m(x) \wedge c(x)]$$

3. Consider the following statement with the real numbers as universe $P(x): x > 0$, $q(x): x^2 \geq 0$, $r(x): x^2 - 3x - 4 = 0$, $s(x): x^2 - 3 > 0$. Determine the truth values of the following quantifiers to the propositions.

i) $\exists x [P(x) \wedge q(x)]$

$$P(x): x > 0$$

$$q(x): x^2 \geq 0$$

for $x=1: P(1): 1 > 0$ T

$x=1: q(1): 1^2 \geq 0$ T

$P(x) \wedge q(x)$ is true

$\exists x [P(x) \wedge q(x)]$ the domain true

ii) $\forall x [P(x) \rightarrow q(x)]$

$$P(x): x > 0$$

$$q(x): x^2 \geq 0$$

for $x=1: P(1): 1 > 0$ T

$x=1: q(1): 1^2 \geq 0$ T

$P(x) \rightarrow q(x)$ is true

$\forall x [P(x) \rightarrow q(x)]$ the domain true

iii) $\forall x [q(x) \rightarrow s(x)]$

$$q(x): x^2 \geq 0$$

$$s(x): x^2 - 3 > 0$$

for $x=1: q(1): 1^2 \geq 0$ T

$x=1: 1^2 - 3 > 0$ F

$q(x) \rightarrow s(x)$ is false

$\forall x [q(x) \rightarrow s(x)]$ the domain false

$$iv) \forall x [r(x) \vee s(x)]$$

$$r(x) : x^2 - 3x - 4 = 0$$

$$s(x) : x^2 - 3 > 0$$

$$\text{for } x=1 : r(1) = 1 - 3 - 4 \neq 0 \quad F$$

$$x=1 : s(1) = 1^2 - 3 > 0 \quad F$$

$r(x) \vee s(x)$ is false

$\forall x [r(x) \vee s(x)]$ the domain false

$$v) \forall x [r(x) \rightarrow p(x)]$$

conjunction (A)

$$r(x) : x^2 - 3x - 4 = 0$$

$$p(x) : x > 0$$

$$\text{for } x=1 : r(x) = 1^2 - 3 - 4 \neq 0 \quad F$$

$$x=1 : p(x) : 1 > 0 \quad T$$

$r(x) \rightarrow p(x)$ is false

$\forall x [r(x) \rightarrow p(x)]$ the domain false

$$vi) \exists x [p(x) \wedge r(x)]$$

disjunction (v)

$$p(x) : x > 0$$

$$r(x) : x^2 - 3x - 4 = 0$$

$$\text{for } x=1 : p(1) : 1 > 0 \quad T$$

$$x=1 : r(1) : 1^2 - 3 - 4 \neq 0 \quad F$$

$p(x) \wedge r(x)$ is false

$\exists x [p(x) \wedge r(x)]$ the domain ~~false~~ true

Rules of Inferences:

Consider the propositions of the statement is $P_1, P_2, \dots, P_n$. compound statement A is a conjunction

the set of propositions are $[P_1 \wedge P_2 \wedge P_3 \dots \wedge P_n] \rightarrow C$

then the argument $P_1 \wedge P_2 \wedge P_3 \dots P_n$ is said to be a premisses and C is a conclusion of the statement.

Valid argument:-

If it is a valid argument $P_1, P_2 \dots P_n$ is a conjunction of $P_1 \wedge P_2 \dots \wedge P_n$.

If the truth values are all its possible situation is true

Consistency of Premises:

1. Inconsistency of premises:

The premisses $P_1, P_2 \dots P_n$ of an argument is said to be inconsistency if their conjunction $P_1, P_2 \dots P_n$ all possible situations are false of an arguments

2. Consistency of premises:

The premisses $P_1, P_2 \dots P_n$ of an argument is said to be consistency if their conjunction $P_1, P_2 \dots P_n$ is true for atleast one possible situation

P1 Test for consistency of premisses $P \wedge (\neg P \wedge q)$

P	q	$\neg P$	$\neg P \wedge q$	$P \wedge (\neg P \wedge q)$
T	T	F	F	F
T	F	F	F	F
F	T	T	T	F
F	F	T	F	F

$\therefore$ The premisses of consistency $P \wedge (\neg P \wedge q)$ is inconsistency

2) ~~$(P \vee q)$~~ and

Test for premisses of consistency $(P \vee q) \wedge (\neg P)$

P	q	$P \vee q$	$\neg P$	$(P \vee q) \wedge (\neg P)$
T	T	T	F	F
T	F	T	F	F
F	T	T	T	T
F	F	F	T	F

$\therefore$ Atleast one possible situation is true then it is the premisses of consistency $(P \vee q) \wedge (\neg P)$ is consistency.

S.NO	Conjunction of Premises	Tautology	Name
1.	$\begin{array}{l} P \\ P \vee q \\ \hline \therefore q \end{array}$	$[P \wedge (P \vee q)] \rightarrow q$	Modulus ponens
2.	$\begin{array}{l} \sim q \\ P \rightarrow q \\ \hline \therefore \sim P \end{array}$	$[(\sim q) \wedge (P \rightarrow q)] \rightarrow (\sim P)$	Modulus Tollens
3.	$\begin{array}{l} P \rightarrow q \\ q \rightarrow r \\ \hline \therefore P \rightarrow r \end{array}$	$[(P \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (P \rightarrow r)$	Hypothesis sollygism
4.	$\begin{array}{l} P \vee q \\ \sim P \\ \hline \therefore q \end{array}$	$(P \vee q) \wedge (\sim P) \rightarrow q$	Disjunctive sollygism
5.	$\begin{array}{l} P \\ \hline \therefore P \vee q \end{array}$	$P \rightarrow (P \vee q)$	Addition rule
6.	$\begin{array}{l} P \wedge q \\ \hline \therefore P \end{array}$	$(P \wedge q) \rightarrow P$	Simplification rule
7.	$\begin{array}{l} P \\ q \\ \hline \therefore P \wedge q \end{array}$	$[(P) \wedge (q)] \rightarrow P \wedge q$	Conjunction rule
8.	$\begin{array}{l} P \vee q \\ \sim P \vee r \\ \hline \therefore q \vee r \end{array}$	$(P \vee q) \wedge (\sim P \vee r) \rightarrow (q \vee r)$	Resolution rule

P1 state which rule of inferences is the bases of following arguments

"If it is below freezing now

If it is below freezing now or raining now

$\therefore$ It is raining now

P is proposition of primeses

If it is below freezing now = P

ly raining. now = Q

$$\begin{array}{l} P \\ P \vee Q \\ \hline \therefore Q \end{array}$$

The valid argument is modulus ponens

$$\therefore [P \wedge (P \vee Q)] \rightarrow Q$$

P2 "If it is below freezing now and raining now

$\therefore$ If it is below freezing now

P is proposition of primeses.

If it is below freezing now = P

raining now = Q

$$\begin{array}{l} P \wedge Q \\ \hline \therefore P \end{array}$$

The valid argument is Simplification rule

$$\therefore [P \wedge Q] \rightarrow P$$

P3 Test whether the following arguments valid or not

"If Sachin hits a century then he gets a free car
Sachin hits a century

$\therefore$ Sachin gets a free car

$$\begin{array}{l} P \rightarrow Q \\ P \\ \hline \therefore Q \end{array}$$

It is not a valid argument

$$[P \rightarrow Q \wedge P] \rightarrow Q$$

$Q \wedge Q \rightarrow 1$
 $Q \wedge \neg Q \rightarrow 0$
 $F \vee 1$
 1

the following

$$[P \Rightarrow P]$$

1930

ive T 3

$$0 \rightarrow 0 \Rightarrow 0$$

... Toller

Modulus Toller

$$Q \Rightarrow P$$

diffn law

ing to the

inverse, inverse

... to be

not P. 113, 114

Inverse: It is inversion of implication

Statement $\sim p \rightarrow \sim q$

Ex: I am not going to school, today is not rainy.

Contrapositive: The negation of converse

Statement $\sim q \rightarrow \sim p$

Ex: Today is not rainy, I am not going to school.

Truth table:

P	q	$p \rightarrow q$	$q \rightarrow p$	$\sim p$	$\sim q$	$\sim p \rightarrow \sim q$	$\sim q \rightarrow \sim p$
T	T	T	T	F	F	T	T
T	F	F	T	F	T	F	T
F	T	T	F	T	F	T	F
F	F	T	T	T	T	T	T

$$\textcircled{1} q \rightarrow p \equiv \sim p \rightarrow \sim q$$

Converse $\equiv$ Inverse

$$\textcircled{2} p \rightarrow q \equiv \sim q \rightarrow \sim p$$

Implication $\equiv$ contrapositive

P: Consider the following propositions with the set of all real numbers as the universe. If the statement

$$P(x): x > 0$$

$$Q(x): x^2 - 3x - 4 = 0$$

Find the truth value of quantify proposition $\forall x [P(x) \rightarrow Q(x)]$ and also write down implication, converse, inverse, contrapositive and find their truth values.

$$P(x): x > 0$$

$$Q(x): x^2 - 3x - 4 = 0$$

$$\forall x [P(x) \rightarrow Q(x)]$$

i) implication :- $P(x) \rightarrow q(x)$

at $x=4$ $P(4): 4 > 0$ T

$q(4): (4)^2 - 3(4) - 4 = 0$ T

$P(x) \rightarrow q(x)$ is true

$\forall x [P(x) \rightarrow q(x)]$ is true

$\therefore$ If for all x is true then $P(x): x > 0$ is true &
 $q(x): x^2 - 3x - 4 = 0$ is also true.

ii) Converse :- $q(x) \rightarrow P(x)$

$\forall x [q(x) \rightarrow P(x)]$ is false

$q(x): x^2 - 3x - 4 = 0$ is false

& $P(x): x > 0$ is false.

iii) Inverse :- $\sim P(x) \rightarrow \sim q(x)$

$\sim P(x): x \leq 0$ F

$\sim q(x): x^2 - 3x - 4 \neq 0$ F

$\sim P(x) \rightarrow \sim q(x)$ is false

$\forall x [\sim P(x) \rightarrow \sim q(x)]$ false

if for all x is false then $\sim P(x): x \leq 0$; $\sim q(x):$
 $x^2 - 3x - 4 = 0$ is false

iv) Contrapositive :- $\sim q(x) \rightarrow \sim P(x)$

$\sim q(x): x^2 - 3x - 4 \neq 0$

$\sim P(x): x \leq 0$

$\sim q(x) \rightarrow \sim P(x)$ is true

$\forall x [\sim q(x) \rightarrow \sim P(x)]$ is true

if for all x is true then $\sim P(x): x \leq 0$,

$q(x): x^2 - 3x - 4 \neq 0$ is true

$$P: q(x): x^2 > 0$$

$$S(x): x^2 - 3 > 0$$

$$\forall x [q(x) \rightarrow S(x)]$$

$$P.T: \forall x P(x) \wedge \forall x q(x) \Leftrightarrow \forall x [P(x) \wedge q(x)]$$

Consider $x = a \in S$

$\Leftrightarrow \{ \text{for all } a \in S \text{ of } P(x) \text{ is true and for all } a \in S \text{ of } q(x) \text{ is true} \}$

$\Leftrightarrow \{ \text{for all } a \in S \text{ of } P(x) \text{ is true } \wedge \text{ for } a \in S \text{ of } q(x) \text{ is true} \}$

$\Leftrightarrow \{ \text{for } a \in S \text{ of } P(x) \wedge q(x) \text{ both are true} \}$

$$\Leftrightarrow \forall x [P(x) \wedge q(x)]$$

$\therefore \forall x P(x) \wedge \forall x q(x) \Leftrightarrow \forall x [P(x) \wedge q(x)]$ is true and $q(x): x^2 - 3x - 4 = 0$ is also true.

****Normal form****

1) C.N.F (conjective Normal Form)

C.N.F = Sum of product of elementary terms

$$(P \vee q) \wedge (\neg P \vee \neg q)$$

① Obtain CNF for (i) $(\neg p \rightarrow r) \wedge (p \rightarrow q)$

$$(\neg p \rightarrow r) \wedge (p \rightarrow q)$$

$$(p \vee r) \wedge (\neg p \vee q) \quad (\because \text{by distributing})$$

$$(p \rightarrow q) \equiv \neg p \vee q$$

$$(\neg p \rightarrow q) \equiv p \vee q$$

$$(p \wedge (\neg p \vee q)) \vee (r \wedge (\neg p \vee q))$$

$$((p \wedge \neg p) \vee (p \wedge q)) \vee ((r \wedge \neg p) \vee (r \wedge q))$$

$$(F \vee (p \wedge q)) \vee ((r \wedge \neg p) \vee (r \wedge q))$$

$$(p \wedge q) \vee (r \wedge \neg p) \vee (r \wedge q)$$

$$(p \vee (r \wedge \neg p)) \wedge [q \vee (r \wedge \neg p) \wedge (r \vee q)]$$

$$(p \vee (r \wedge \neg p)) \wedge [q \vee (r \wedge \neg p) \wedge (r \vee q)]$$

$$((p \vee r) \wedge (p \vee \neg p)) \wedge [(q \vee r) \wedge (q \vee \neg p) \wedge (r \vee q)]$$

$$((p \vee r) \wedge T) \wedge ((q \vee r) \wedge (q \vee \neg p) \wedge (r \vee q))$$

By commutative law

$$(P \vee \neg P) \wedge (Q \vee \neg Q) \wedge (R \vee \neg R)$$

$$\textcircled{2} \text{ C.N.F } [P \rightarrow (P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P)]$$

P	Q	$P \rightarrow Q$	$P \rightarrow (P \rightarrow Q)$	$\neg P$	$\neg Q$	$\neg Q \vee \neg P$
T	T	T	T	F	F	F
T	F	F	T	F	T	T
F	T	T	T	T	F	T
F	F	T	F	T	T	T

$$\neg(\neg Q \vee \neg P) \quad [P \rightarrow (P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P)]$$

T	T
F	F
F	F
F	F

A-condition
T-true
F-negative

Note: NO. of possibilities
for false

$$(P \vee \neg Q) \wedge (\neg P \vee Q) \wedge (\neg P \vee \neg Q)$$

$$\pi(1,2,3) = (P \vee \neg Q) \wedge (\neg P \vee Q) \wedge (\neg P \vee \neg Q)$$

possibility of false
Conjunctive normal form (CNF): A formula which is equivalent to given formulas and which consist of Product of elementary sum's is called CNF

→ It is denoted by ' π '

Applications:-

i. It is the formula contains $\Rightarrow$ (or) $\Leftrightarrow$, connects, replace them with equivalent formula and, Or, negation.

ii. The negation is applied to the formula replaced by De Morgan's law.

P) obtain by $P \wedge (P \rightarrow Q)$

$$P \rightarrow Q \equiv \neg P \vee Q$$

$P \wedge (\neg P \vee Q)$ by Distributive

$(P \wedge \neg P) \vee (P \wedge Q)$ by negation

$$(\neg P \vee (P \wedge Q))$$

$$(P \wedge Q)$$

$$P) [P \rightarrow (Q \wedge R) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))]$$

$$[P \rightarrow (Q \wedge R) \wedge (\neg P \rightarrow (\neg Q \wedge \neg R))] \quad \begin{array}{l} P \rightarrow Q \equiv \neg P \vee Q \\ \neg P \rightarrow Q \equiv P \vee Q \end{array}$$

$(\neg P \vee (Q \wedge R)) \wedge (P \vee (\neg Q \wedge \neg R))$ by distributive

$$(\neg P \vee Q) \wedge (\neg P \vee R) \wedge (P \vee \neg Q) \wedge (P \vee \neg R)$$

$$(P \rightarrow Q) \wedge (P \rightarrow R) \wedge (P \rightarrow \neg Q) \wedge (P \rightarrow \neg R)$$

Principle's of CNF :-

Principle CNF do not connected in the p, q, r then

connective to $P \wedge \neg P, q \wedge \neg q, r, \neg r$

P) obtain by PCNF $(\neg P \rightarrow r) \wedge (q \leftrightarrow P)$

$$P \rightarrow Q \equiv \neg P \vee Q$$

$$(P \vee q) \wedge (q \rightarrow P) \wedge (P \rightarrow q)$$

$$\neg p \rightarrow q \equiv P \vee q$$

$(P \vee r) \wedge (\neg q \vee P) \wedge (\neg P \vee q)$ by Demorgan law

$$(P \vee r) \vee (q \wedge \neg q) \wedge (\neg q \vee P) \vee (r \wedge \neg r) \wedge (\neg P \vee q) \vee (r \wedge \neg r)$$

$$((P \vee r) \vee q) \wedge ((P \vee r) \vee \neg q) \wedge ((\neg q \vee P) \vee r) \wedge (\neg q \vee P \vee \neg r) \wedge ((\neg P \vee q) \vee r) \wedge (\neg P \vee q \vee \neg r) \quad \text{by associative law}$$

$$(P \vee q \vee r) \wedge (P \vee \neg q \vee r) \wedge (P \vee \neg q \vee r) \wedge (P \vee \neg q \vee \neg r) \wedge (\neg P \vee q \vee r) \wedge (\neg P \vee q \vee \neg r)$$

$$P) (P \rightarrow R) \wedge (P \rightarrow Q) \Leftrightarrow (P \rightarrow R) \wedge (\neg P \rightarrow Q) \wedge (Q \rightarrow \neg P)$$

$$\stackrel{\text{RHS}}{=} (P \rightarrow R) \wedge (\neg P \rightarrow Q) \wedge (Q \rightarrow \neg P)$$

$$P \rightarrow Q \equiv \neg P \vee Q$$

$$\neg P \rightarrow Q \equiv P \vee Q$$

$$(\neg P \vee R) \wedge (P \vee Q) \wedge (Q \vee \neg P)$$

$$(\neg P \vee R) \vee (Q \wedge \neg Q) \wedge (P \vee Q) \vee (R \wedge \neg R) \wedge (Q \vee \neg P) \vee (R \wedge \neg R)$$

$$(\neg P \vee R \vee Q) \wedge (\neg P \vee R \vee \neg Q) \wedge (P \vee Q \vee R) \wedge (P \vee Q \vee \neg R) \wedge (Q \vee \neg P \vee R) \wedge (Q \vee \neg P \vee \neg R)$$

$$(\neg P \vee Q \vee R) \wedge (\neg P \vee \neg Q \vee R) \wedge (P \vee Q \vee R) \wedge (P \vee Q \vee \neg R) \wedge (Q \vee \neg P \vee R) \wedge (Q \vee \neg P \vee \neg R)$$

$$\stackrel{\text{LHS}}{=} (P \rightarrow R) \wedge (P \rightarrow Q)$$

$$(\neg P \vee R) \wedge (\neg P \vee Q)$$

$$(\neg P \vee R) \vee (Q \wedge \neg Q) \wedge (\neg P \vee Q) \vee (R \wedge \neg R)$$

$$((\neg P \vee R) \vee Q) \wedge ((\neg P \vee R) \vee \neg Q) \wedge ((\neg P \vee Q) \vee R) \wedge ((\neg P \vee Q) \vee \neg R)$$

$$((\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R) \wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R))$$

Disjunctive normal form (DNF): possibility of true

A formula in which consist of the sum of elementary product is said to be DNF

→ It is denoted by 'Σ'

→ It is possible situations of true

P) Obtain DNF $\neg(P \rightarrow (Q \wedge R))$

$\neg(P \rightarrow (Q \wedge R))$ by demorgan law

$$\neg(\neg P \vee (Q \wedge R))$$

$P \wedge (\neg Q \vee \neg R)$ by distributive law

$$(P \wedge \neg Q) \vee (P \wedge \neg R)$$

$$P \rightarrow Q \equiv \neg P \vee Q$$

$$\neg P \rightarrow Q \equiv P \vee Q$$

P	Q	R	$(Q \wedge R)$	$P \rightarrow (Q \wedge R)$	$\neg(P \rightarrow (Q \wedge R))$
T	T	T	T	T	F 0
T	T	F	F	T	F 1
T	F	T	F	T	F 2
T	F	F	F	T	F 3
F	T	T	T	F	T - 4
F	T	F	F	T	F 5
F	F	T	F	T	F 6
F	F	F	F	T	F 7

$$\Rightarrow \Sigma(4) = (\neg P \wedge Q \wedge R)$$

* P) Obtain the PDNF and PCNF to the given statement

$$(\neg P \vee \neg Q) \rightarrow (P \leftrightarrow Q)$$

P	Q	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$	Total
T	T	F	F	F	T	T	T	F 0
T	F	F	T	T	T	F	F	T 1
F	T	T	F	T	F	T	F	T 2
F	F	T	T	T	T	T	T	T 3

$$P.C.N.F = (P \vee Q) \wedge (\neg P \vee \neg Q)$$

$$P.D.N.F = (P \wedge \neg Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q)$$

② find DNF $P \rightarrow (P \rightarrow Q) \wedge \neg(Q \vee \neg P)$

P	Q	$P \rightarrow Q$	$P \rightarrow (P \rightarrow Q)$	$\neg P$	$\neg Q$	$(Q \vee \neg P)$	$\neg(Q \vee \neg P)$	total
T	T	T	T	F	F	F	T	T
T	F	T	T	F	T	T	F	F
F	T	F	T	T	F	T	F	F
F	F	T	F	T	T	T	F	F

3) P.T $(P \rightarrow Q) \rightarrow R$ and $(P \wedge \neg Q) \vee R$ is logically equivalence

Sol: $(P \rightarrow Q) \rightarrow R \equiv (P \wedge \neg Q) \vee R$

$\neg(P \rightarrow Q) \vee R$ By Demorgan's

$\neg(\neg P \vee Q) \vee R$

$(P \wedge \neg Q) \vee R$ by negation law

LHS $(P \vee R) \wedge (\neg Q \vee R)$ by Distributive

RHS $(P \wedge \neg Q) \vee R$ by Dis

$(P \vee R) \wedge (\neg Q \vee R)$

4) construct truth-table to the given statement

$(P \wedge Q) \vee (\neg P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$

5) Obtain the PCNF $(\neg P \rightarrow Q) \wedge (Q \leftrightarrow P)$ with out using truth table

$(\neg P \rightarrow Q) \wedge (Q \leftrightarrow P)$

$(\neg P \rightarrow Q) \wedge (Q \rightarrow P) \wedge (P \rightarrow Q)$

$(P \vee Q) \wedge (\neg Q \vee P) \wedge (\neg P \vee Q)$

6) show that $(P \vee Q) \wedge (\neg P \wedge (\neg P \wedge Q)) \Leftrightarrow (\neg P \wedge Q)$

$(P \vee Q) \wedge (\neg P \wedge \neg P) \wedge Q$ by associative law

$(P \vee Q) \wedge (\neg P \wedge Q)$

$(P \wedge \neg P \wedge Q) \vee (Q \wedge \neg P \wedge Q)$

$(F \wedge Q) \vee (\neg P \wedge Q \wedge Q)$

$F \vee (\neg P \wedge Q)$

$\neg P \wedge Q$

7) Obtain PCNF and PDNF $P \vee (\neg P \rightarrow (Q \vee (\neg Q \rightarrow R)))$

8) show that $\neg(P \wedge Q), (\neg Q \vee R), \neg R \Rightarrow \neg P$ is valid
valid argument or not?

additional Premises P

1

P

Rule - P

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2 $\neg p \vee \neg q$

[1,2](3) $\neg q$

4 $(\neg q \vee r)$

[1,2,4](5) $\times$

6 $\neg r$

[1,2,4,6](7) F

Rule-P

Rule-T (1)(3) $[P, \neg p \vee \neg q]$
Modulus Ponens

Rule-P

Rule-T (3)(4) $[\neg q, \neg q \vee r]$
Modulus Ponens

Rule-P

$\therefore$ The new premises P is contradiction to valid argument

$\therefore$ It is not valid

P) Test the valid argument

All men are mortal

Sachin is a man

$\therefore$ Sachin is a mortal

S is set of characters

$x = a \in S$

$P(x)$: men are mortal

$q(x)$: Sachin is a man

$\forall x p(x)$

$q(x)$

$\therefore p(a)$

$\therefore$ It is valid argument

P) Find the following arguments

All students are engineering students then

NO engineering student is bad in studies

Anil is not bad in studies

$\therefore$ Anil is an engineering student

$P(x)$: student are engineering

S is set of characters

$x = a \in S, a = \text{Anil}$

$\forall x [P(x) \rightarrow \neg q(x)]$

$\neg q(a)$

$\therefore P(a)$

Methods of proofs and strategy:-

strategy includes selecting a proof of methods and then successfully constructing an argument step by step to find a proof of methods.

Theorem ①: S.T n is odd then n^2 is also odd number?

Proof: Consider $n = 2k + 1$

for $k = 0$

$$n = 2(0) + 1$$

$$n = 1$$

for $k = 1$

$$n = 2(1) + 1$$

$$n = 3$$

$\therefore n$ is odd number

$$n^2 = (2k + 1)^2$$

$$n^2 = 4k^2 + 4k + 1$$

for $k = 0$

$$n^2 = 1$$

for $k = 1$

$$n^2 = 9$$

n^2 is odd

$\therefore n$ is odd then n^2 is also odd number

Theorem 2: S.T the two irrational numbers to the given expression if x and y is irrational number then x^y is a rational number

Proof:- Consider,

x is irrational number $= (\sqrt{2})^{\sqrt{2}}$

y is irrational number $= \sqrt{2}$

$$x^y = [(\sqrt{2})^{\sqrt{2}}]^{\sqrt{2}}$$

$$x^y = (\sqrt{2})^2$$

$$x^y = 2$$

$\therefore x$ & y are irrational then x^y is a rational number

3. Proof that the sum of two rational number is
 r, s . if r, s are rational numbers then $r+s$ is
 also a rational number

Consider, $r = p/q$

$$s = u/v$$

$$r+s = p/q + u/v$$

$$r+s = \frac{pv + qu}{qv}$$

$\therefore r$ & s are rational number

$r+s$ is also a rational number

4. S.T $|xy| = |x| \cdot |y|$ then x & y are real numbers it
 is absolute value of p & q

Proof:- $|xy| = |(+x)| \cdot |(+y)|$; $x > 0, y > 0$

$$|xy| = |(+x)| \cdot |(-y)|$$
; $x > 0, y < 0$

$$|xy| = |(-x)| \cdot |(+y)|$$
; $x < 0, y > 0$

$$|xy| = |(-x)| \cdot |(-y)|$$
; $x < 0, y < 0$